DA-HELLO: State-Carrier-Dependent HELLO Scheduling for Q-Learning Routing in Underwater Sensor Networks

Xuefei Ma, Ghulam Ali, Zhongwei Shen, Insha Hassan, Muhammad Aon, Qingling Liu, Haifeng Zhu, and Ying Cui

***Abstract*—Q-learning routers for underwater sensor networks rank forwarders using held neighbour state, yet the cost of keeping that state current depends on how the state travels. We study this dependence with DA-HELLO, a configurable HELLO scheduler for a host router. A node sends a HELLO when it first decodes a never-recorded sender, when its candidate membership or void status changes, and, optionally, on a slow refresh; a local score-change test is also included. We evaluate it on two Q-learning routers, MARV and QELAR, each with state carried in data headers and in a constructed HELLO-only variant, in static, drifting, and node-loss scenarios. Comparisons are paired over 40 seeds, statements and margins were fixed before the runs, and idle listening is counted in energy. When state travels only in HELLOs, DA-HELLO delivers 2.1 to 3.0 percentage points more than a jittered 120 s periodic schedule on MARV, with 43 to 53% fewer control bytes and 8 to 10% lower energy per bit. On QELAR it uses 45 to 52% fewer bytes and gains 5.0 points in the static scenario, but is 2.2 and 1.7 points lower in the drifting ones, so non-inferiority at 3 points is not shown. When state is carried in data headers, DA-HELLO matches sending no HELLOs, whereas periodic and Trickle schedules lose delivery. The score-change test shows no demonstrated benefit. The configuration differs by host class, and the results come from simulation of two re-implemented routers.**

***Index Terms*—HELLO scheduling, neighbour-state maintenance, Q-learning routing, state carriage, underwater acoustic sensor networks.**

I. Introduction

THE Internet of Underwater Things (IoUT) links sensors, vehicles, and gateways into acoustic multi-hop meshes for environmental observation, resource exploration, and maritime awareness [1]–[4]. Acoustic links offer little bandwidth, long propagation delay, and a time-varying multipath channel [1], [3], [5], and nodes drift with currents or are carried by vehicles, so a node's neighbourhood keeps changing [2], [6]–[8]. Reinforcement-learning routers have become a common response: each node learns a forwarding policy from the state of its neighbours, such as value estimates, residual energy, depth, and link quality [9]–[13].

Such a router acts on *held* neighbour state, not on the current state of the network, and the way that state reaches the receiver differs between routers. In some, it travels in data headers, so a node that forwards traffic advertises itself at no extra cost, while a node off the forwarding paths advertises nothing. In others, it travels only in HELLO or beacon frames, and some use both (Section II). The two cases pose different maintenance problems. If state is carried in headers, an additional HELLO may add little. If it is carried only in HELLOs, a missing or old record can make a candidate invisible or wrongly ranked, and the packet is delayed, detoured, or lost. In both cases, every HELLO costs scarce acoustic airtime and energy, including idle listening and the preamble of a short frame.

The literature handles when to re-advertise in three ways: a fixed timer, a timer adapted to mobility [14]–[16], or a trigger on a void or topology change [17], [18]. Receive-driven replies to a newly overheard neighbour are known from adaptive beaconing in ad hoc networks [15], and void-status beacons from void-handling protocols [17]. What remains open for underwater Q-learning routers is an evaluation that holds the routing logic fixed and varies only the maintenance policy across routers whose state travels in different ways. Most underwater studies also assert that control overhead is negligible without measuring it [19]–[21], and most energy results leave idle listening out [18]–[20].

We address this with DA-HELLO, a configurable HELLO scheduler that sits beside a host router and needs only the host's neighbour query and broadcast primitive. A node sends a HELLO on a first decode of a never-recorded sender, on a change of candidate membership or void status, and, optionally, on a slow refresh; a local score-change test is also included. We evaluate it on two Q-learning routers, MARV and QELAR, and treat LQARP and SAQR as pilot hosts. Our contributions are:

1) **Problem.** We formulate neighbour-state maintenance for Q-learning routers as a choice that depends on how state reaches the receiver, in data headers or only in HELLOs, and pose it as minimising control overhead under a bound on the loss of delivery ratio. Without HELLOs, a HELLO-only host collapses to a delivery ratio of 0.06 to 0.18, whereas on header hosts periodic and Trickle HELLOs reduce delivery.

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2) **Method.** DA-HELLO is a configurable scheduler that needs only the host's neighbour query and broadcast primitive. It sends on the first decode of a never-recorded sender, on a change of candidate membership or void status, and optionally on a slow refresh; a local score-change test is also included. We state the evaluated semantics exactly and do not claim that each HELLO is necessary for a receiver's decision.
3) **Evaluation.** Statements, margins, and seeds were fixed before the runs; comparisons are paired over 40 seeds on MARV and QELAR in header and HELLO-only form, and energy includes idle listening. On HELLO-only MARV, DA-HELLO delivers 2.1 to 3.0 percentage points more than a jittered 120 s periodic HELLO with 43 to 53% fewer control bytes and 8 to 10% less energy per bit. On HELLO-only QELAR it uses 45 to 52% fewer bytes and gains 5.0 points in the static scenario, but is 2.2 and 1.7 points lower in the drifting ones, so non-inferiority at 3 points is not shown. On header hosts it matches sending no HELLOs. The slow refresh carries delivery on HELLO-only hosts, first-contact replies harm header QELAR, and the score-change test has no demonstrated benefit. Limits of the simulation record are disclosed.

The rest of the article is organised as follows. Section II reviews related work and the gap. Section III gives the preliminaries. Section IV defines the problem and the design. Section V presents the simulation settings and results, and Section VI concludes.

II. Related Work

This section reviews the literature with a focus on how a node learns the state of its neighbors and what determines when it re-advertises its own. Table I summarizes this for the learning-based UWSN routers closest to our setting; the remaining subsections cover adaptive HELLO schemes in Q-learning routers for other mobile networks, change-driven control messages in ad hoc routing and vehicular standards, adaptive beaconing and Trickle, and the estimation and information-freshness literature.

A. Learning-Based UWSN Routing and Neighbor-State Maintenance

Reinforcement-learning routers for underwater networks differ in their learning rule, ranging from tabular Q-learning with an energy-aware reward [9] to channel-aware distributed learning [22], service-aware rewards [10], void recovery [23], multi-agent formulations [11], [12], and rewards built on multipath-aware link quality [13]. They share one requirement: a forwarder needs its neighbors' values (value estimates, residual energy, depth, or link quality) at decision time. The protocols meet it in three ways.

The first is to carry the state in data packets and learn it by overhearing, as in QELAR, CARMA, DROR, SAQR, and LQARP. This costs no control traffic, but a node that is not on a forwarding path advertises nothing. MARV makes the point explicit: besides the header fields, each node broadcasts a control packet "periodically" so that neighbors learn its state "even if they do not have packets to forward" [11]. The second is a timer. Chang *et al.* have every node beacon at the start of each time slot [24]; MARV and CLORP likewise rely on a periodic control message [11], [12]. In these designs the interval is a fixed constant, and in several of them it is not reported [11], [12], [24]. Nothing links it to how quickly the advertised state is changing.

The third adapts the timer. QLAR predicts neighbor positions and velocities with an extended Kalman filter and sets the HELLO interval to a fraction τ of the shortest predicted link-expiration time among neighbors [14]. It is the closest UWSN precedent to our work, and we run its link-expiration-scaled interval as a baseline (Section V). Its connectivity model is a fixed communication range, so link expiry is a purely geometric event, and it remains a timer rather than a test of whether the advertised change matters to any neighbor's decision.

Mobility enters the UWSN literature in other ways as well. Chang *et al.* model passively drifting sensors but keep the per-slot beacon [24]; cooperative multi-AUV collection plans vehicle paths with deep Q-learning [6]–[8]; an AUV can be dispatched to repair isolated sub-networks [25]; acoustic–radio meshes switch media opportunistically [26]; routes, power, and link schedules have been optimized jointly [27], [28]; and spatial-temporal graphs record neighbor relations over periodic intervals for multi-UUV routing [29]. In these designs motion affects where a vehicle goes or which medium is used; none uses it to decide when a neighbor should be told.

B. Adaptive HELLO Intervals in Q-Learning and Ad Hoc Routing

Outside UWSNs, several Q-learning FANET/MANET routers adapt the HELLO period to speed, neighbor churn, or predicted link lifetime [30]–[34]. These schemes set *how often* to advertise from kinematics or topology change rates; they do not test whether a given attribute change would alter a learned ranking at a neighbor. Therefore, rate adaptation alone does not address decision relevance for Q-learning hosts.

C. Change-Driven Control Messages

Triggered updates appear in classical ad hoc routing and neighbor discovery: DSDV triggered updates, TBRPF next-hop-relevant reporting, NHDP responsive HELLOs, and ETSI CAM rules keyed on kinematics [35]–[38]. The common pattern is to advertise when a link, metric, or trajectory changes beyond a threshold. However, the trigger is defined on topology or kinematics rather than on whether a *learned* score would change a neighbor's next-hop choice. Therefore, a decision-relevance test for Q-learning hosts remains open.

D. Adaptive Beaconing and Trickle

Terrestrial networks adapt the beacon rate to speed or neighbor turnover [39], [40]. Trickle doubles its interval while neighbors are consistent and resets on inconsistency [41], [42]; RPL applies it to routing control and resets on routing-state

TABLE I
HOW LEARNING-BASED UWSN ROUTERS KEEP NEIGHBOR STATE CURRENT ("N.P." = NOT PUBLISHED). LAST ROW: DA-HELLO.

Protocol	Learning	How neighbor state is disseminated	Control-message interval	Rule for re-advertising
QELAR [9]	Q-learning (model-based)	Carried in data packets	none	Data-driven; no control message
CARMA [22]	Distributed RL	Header fields and overhearing; 6-byte HELLO for discovery	n.p.	Data-driven
DROR [23]	Q-learning + opportunistic	Header fields; sink broadcast on delivery	none	Data-driven
SAQR [10]	Q-learning	Carried in data packets	none	Data-driven
Chang *et al.* [24]	RL	Beacon from every node	One per time slot (slot length n.p.)	Fixed timer
MARV [11]	Multi-agent RL	Header fields plus a periodic control packet	n.p.	Fixed timer
CLORP [12]	Multi-agent RL	HELLO message	n.p.	Fixed timer
QLAR [14]	Q-learning + EKF	Periodic HELLO with position, velocity, and V-value	$\tau \min_j T_{\mathrm{LET}}$, $\tau = 0.3$	Timer scaled by predicted link-expiration time under a fixed-range model
LQARP [13]	Q-learning	Header fields (V-value, neighbor count); position and energy from one initialization broadcast	none	Data-driven
DA-HELLO (this work)	Q-learning hosts (host-parametric)	HELLO on demand; one broadcast at first contact	Event-driven	Change could alter a neighbor's candidate set or order

inconsistency [43]. These schemes answer *how often* to beacon given mobility or churn, not whether a given change would alter a learned candidate order. We therefore use periodic timers and a tuned Trickle as baselines.

E. Event-Triggered Estimation and the Value of Updates

Event-triggered estimation transmits an update when a prediction residual or innovation exceeds a threshold [44]–[48]. Li *et al.* use media-specific triggers for a hybrid underwater network [49]; Kalman and Markov models have been compared for predicting acoustic SNR from traces [50]. In all of these works the trigger protects an estimation error. Age-of-information scheduling instead asks how often a source should update so that the receiver's information stays fresh [51], [52]; value-of-information and goal-oriented communication transmit an update only when it matters to the receiver's goal [53]–[55]. Our trigger is a concrete instance of this goal-oriented principle in which the receiver's goal is its next-hop choice; it does not use an estimation-error trigger.

F. Positioning and Research Gap

The principle "advertise only what can change a neighbor's routing decision" is therefore not new: it appears in DSDV triggered updates, TBRPF's next-hop-relevant reporting, NHDP responsive HELLOs, RPL's inconsistency reset, QFAGR's hole broadcasts, TARRAQ's event-driven sensing interval, and goal-oriented updating. On the campaigned hosts, the realized DA-HELLO traffic under locked configurations is dominated by first-contact replies, membership (void-status) events, and optional slow refresh rather than by a continuously firing rank test. What remains open, and what this paper contributes, is narrower. First, a formal definition of decision relevance for *learned* scores: a change is relevant if it alters whether a node is a candidate of some neighbor or flips the sign of a score gap to a co-candidate (Section III), which no shortest-path or threshold rule captures for Q-learning hosts. Second, a local, host-parametric estimator of that test that a sender can evaluate from the information it is allowed to hold, with the host entering only through its candidate test and score. Third, a cross-host evaluation of HELLO scheduling for Q-learning UWSN routers with different control planes, against tuned adaptive baselines that include QLAR's link-expiration timer, under one energy-accounting rule that includes idle listening, and on a shared acoustic medium with a contention MAC, half-duplex operation, and heard-only neighbor discovery.

III. SYSTEM MODEL AND PROBLEM FORMULATION

DA-HELLO acts on neighbour state that a node holds, so this section defines what a node can know and how that state reaches it. It introduces the two state carriers that organise the rest of the paper and contains no parameter values; those are given in Section V.

A. Network and Mobility

A set of relays $\mathcal{S} = \{s_1, \ldots, s_N\}$ and one fixed sink s_0 share an acoustic medium in a three-dimensional volume (Fig. 1). Each relay s_i has a position $\mathbf{p}_i(t)$, a depth $z_i(t)$, a residual energy $e_i(t)$, a queue, learning variables, and timers. Sensing nodes generate packets, and every packet is addressed to the sink and relayed hop by hop.

Relays drift with the water. The velocity of relay s_i follows a Gauss–Markov process with memory ζ around a position-dependent mean flow $\boldsymbol{\mu}(\mathbf{p})$,

$$\mathbf{v}_i(t+\Delta) = \zeta\, \mathbf{v}_i(t) + (1-\zeta)\, \boldsymbol{\mu}(\mathbf{p}_i) + \sqrt{1-\zeta^2}\, \boldsymbol{\Sigma}\, \mathbf{w}_i, \quad (1)$$

where $\mathbf{w}_i$ is standard Gaussian noise and $\boldsymbol{\Sigma}$ sets the per-axis deviation. The mean flow models a mesoscale eddy as a counter-clockwise tangential flow around a centre $\mathbf{c}$ with radius R and peak speed V,

$$\boldsymbol{\mu}(\mathbf{p}) = V\, \min\!\Big(\tfrac{\rho}{R}, 1\Big)\, \hat{\mathbf{e}}_\theta, \qquad \rho = \|\mathbf{p}_{xy} - \mathbf{c}\|, \quad (2)$$

so nodes at different radii move differently and links break and form. The sink is stationary. A relay that leaves the volume is removed for the rest of the run. Static scenarios correspond to $V = 0$ and $\boldsymbol{\Sigma} = \mathbf{0}$.

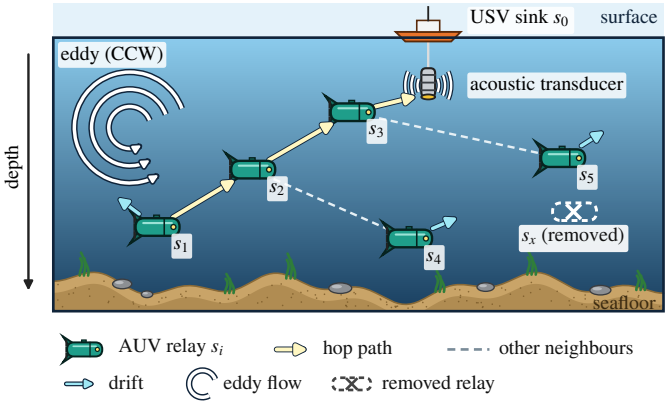


Fig. 1. Network model (side view). Relays drift under an eddy and Gauss–Markov flow, so decodable links form and break; packets are forwarded hop by hop to the fixed sink. A relay that leaves the volume is removed. Each relay forwards using *held* records of its neighbours, whose upkeep is the subject of this work.

B. Channel, Frames, and Energy

Only the medium evaluates propagation and reception. The signal-to-noise ratio of a frame sent over a distance d is

$$\mathrm{SNR}(d) = \mathrm{SL} - 10\kappa \log_{10} d - \alpha(f_c)\, d - N_0, \qquad (3)$$

with source level SL, spreading exponent κ, Thorp absorption $\alpha(f_c)$ at carrier frequency f_c, and ambient noise level N_0, all in decibels. Frames use non-coherent binary frequency-shift keying, for which the bit error rate is $p_b = \frac{1}{2}\exp(-\gamma_b/2)$ with $\gamma_b = \mathrm{SINR} \cdot B/R_b$, and the frame error rate is $1-(1-p_b)^{\ell_f}$ for a frame of ℓ_f bits. Concurrent frames interfere, and a half-duplex node cannot receive while it transmits. A frame of ℓ_f bits occupies the channel for

$$T_f = T_{\mathrm{pre}} + \ell_f / R_b, \qquad (4)$$

so a short control frame is dominated by its preamble T_{pre}. A node draws the powers P_{tx}, P_{rx}, and P_{idle} in the transmit, receive, and idle-listening states, and it listens continuously. A control frame therefore costs $T_f(P_{\mathrm{tx}} - P_{\mathrm{idle}})$ at the sender and $T_f(P_{\mathrm{rx}} - P_{\mathrm{idle}})$ at each receiver above the idle baseline, and it also consumes airtime that data frames could use.

C. Locally Available Information

A node s_i learns of a neighbour s_j only by decoding a frame from it. Let $b_{ij}(t)$ be the latest time at which s_i accepted a state-bearing frame from s_j, with $b_{ij} = -\infty$ if none. A record is *fresh* when

$$L_{ij}(t) = \mathbf{1}\{b_{ij}(t) > -\infty,\ t - b_{ij}(t) \le T_{\mathrm{nb}}\}, \qquad (5)$$

and only fresh records can form candidates. At time t, the routing logic and the HELLO policy of s_i may use only: its own state and timers; the fields of frames it has decoded, as written by the sender at transmission start; the received SNR and decode time of those frames; held records built from them, possibly stale; a-priori constants such as the PHY and MAC settings and the fixed sink position; and the outcome of its own transmissions. A node never reads another node's true position, depth, energy, or neighbour set.

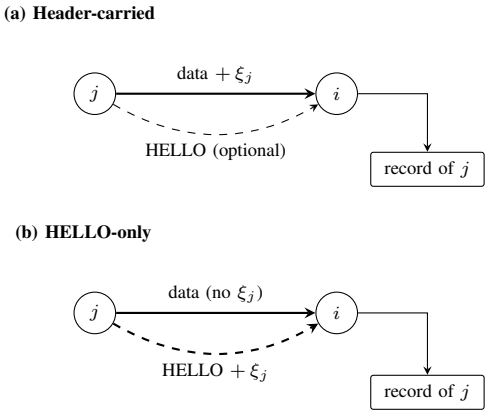


Fig. 2. The two state carriers. (a) Header-carried: a data frame writes the sender's attributes into the receiver's held record; a HELLO is optional. (b) HELLO-only: data frames write nothing, and a record exists only after a HELLO or beacon has been decoded, so maintenance determines whether a neighbour can be ranked at all.

D. State Carriers and Host Interface

Each node s_j has an advertised attribute vector $\xi_j(t)$ that its neighbours need, such as a value estimate, residual energy, depth or position, and a void flag. A frame carries a subset of it. We distinguish two *state carriers* (Fig. 2). In the *header-carried* regime, data frames carry the attributes, so a forwarding node advertises itself at no extra control cost, and a HELLO supplements this for nodes off the forwarding paths. In the *HELLO-only* regime, data frames carry none, and a held record is written only by HELLO or beacon frames. Real protocols may mix the two. In Section V, the HELLO-only regime is a constructed variant of each host, not a reproduction of a published protocol.

A host router h is described by an eligible candidate set $\mathcal{I}_i^h(t)$ of node s_i, built from fresh records, and a forwarding output $\mathcal{R}_h$ that orders the candidates by a score $\sigma_i(j)$ and selects a next hop. The scheduler of Section IV uses the host through three primitives only: the candidate set and held records of the node itself, its own advertised vector, and a broadcast of a control frame. It does not use the host's learning rule. The hosts differ in their eligibility rules: in MARV a candidate has a fresh record, a held depth shallower than the sender, and no excluding void flag, and candidates are ordered by updated Q values; in QELAR a candidate has a fresh record and passes an admission test on its held SNR, without a depth restriction, and the largest stored Q value selects the next hop.

E. Held State and Its Staleness

Between two accepted frames from s_j, node s_i holds a zero-order copy $\hat{\xi}_{ij}$ of the advertised vector, while the true attributes $\xi_j(t)$ keep evolving with motion, energy use, and learning. The copy is wrong for s_i's purposes only if the error changes what s_i does: whether s_j is a candidate, how it ranks against another candidate, or whether s_i can see s_j at all. A record that is merely old is not necessarily wrong. This distinction, which Section IV-A formalises, is what separates deciding when to re-advertise from running a timer.

F. Metrics and Accounting

For a run r, let G_r be the number of application packets generated in the analysis interval $[t_w, t_g)$ after a warm-up t_w, and D_r the number of those packets that reach the sink. The run-level delivery ratio and energy per delivered bit are

$$p_r = \frac{D_r}{G_r}, \qquad e_r = \frac{E_r}{\ell\, D_r}, \tag{6}$$

where ℓ is the data payload length and E_r is the modem energy of all nodes accumulated over $[t_w, t_g + t_d]$, which includes a drain period t_d after the last generation. It covers transmit, receive, and idle-listening states, so idle energy is always counted, and it excludes propulsion and other platform loads. A relay that is removed stops generating packets, but packets it already offered stay in G_r. The packet cohort and the energy interval are therefore different intervals, and we do not call either of them "the window".

The primary summaries average run-level ratios, $\bar{p} = \mathbb{E}[p_r]$ and $\bar{e} = \mathbb{E}[e_r]$, over seeds. They differ from pooled estimates,

$$\hat{p}_{\mathrm{pool}} = \frac{\sum_r D_r}{\sum_r G_r}, \qquad \hat{e}_{\mathrm{pool}} = \frac{\sum_r E_r}{\ell \sum_r D_r}, \tag{7}$$

and one cannot be reported as the other. When $D_r = 0$, e_r is undefined, not zero, and the run's energy still belongs in the pooled numerator. The control overhead is

$$O_{\mathrm{ctrl}} = \frac{1}{t_g} \sum_{f \in \mathcal{F}} \frac{\ell_f}{8}, \tag{8}$$

where $\mathcal{F}$ is the set of HELLO and beacon frames that start in $[0, t_g]$, with ℓ_f in bits. It charges the initial announcements and the sink beacon and excludes frames that start during the drain. The control count, the packet cohort, and the energy interval have deliberately different definitions.

IV. Problem Definition and DA-HELLO Design

A HELLO is useful only if it changes what a receiver does, yet a sender cannot see its receivers' state. This section states that as a scheduling problem and then gives DA-HELLO, a causal heuristic that approximates it from the sender's own history. Only the scheduler is described here; the host routers and the state carriers are those of Section III.

A. Problem Definition

Let $U_{j,k} \in \{0, 1\}$ indicate that node s_j requests a HELLO at decision epoch k of period Δt. A request can be queued, delayed, or dropped by the MAC, and a transmitted HELLO can collide or fall below the decoding threshold, so a request neither guarantees a transmission nor refreshes any particular receiver. Between frames, every receiver holds a zero-order copy of the advertised vector ξ_j, while the true attributes keep evolving.

Not every change matters to a receiver. Let m_j be an advertisement of s_j and $\mathcal{U}_h(X_i^h, m_j, y_{ij})$ the host's processing of its successful reception at s_i, where X_i^h holds the held records and learning state and y_{ij} is the reception metadata. For a diagnostic recipient set $\mathcal{C}_j$, define

$$D_j = \mathbf{1}\big\{\exists\, i \in \mathcal{C}_j :\ \mathcal{R}_h(X_i^{h,+}; \kappa_i) \neq \mathcal{R}_h(X_i^h(t^-); \kappa_i)\big\}, \qquad X_i^{h,+} = \mathcal{U}_h\big(X_i^h(t^-), m_j, y_{ij}\big), \tag{9}$$

that is, the advertisement matters if some receiver's forwarding output $\mathcal{R}_h$ for a packet context κ_i would change. Missing and expired records are included in the hypothetical reception, so a first contact is relevant because a node that has never recorded s_j cannot rank it. Candidacy changes and sign changes of the score gap $g_{ijc} = \sigma_i(j) - \sigma_i(c)$ to a co-candidate s_c are the two ways the output can change. D_j is a receiver-specific offline *diagnostic*: it needs the receivers' state, so it cannot be an input to a node, and it was not measured in our campaign. An update can also be useful later when the immediate output is unchanged, and a changed output can be harmful, so D_j does not define an optimal schedule.

Let $B(\pi)$ be the control bytes sent under a HELLO policy π and $\mathrm{PDR}(\pi)$ the delivery ratio of a fixed host router. With a reference policy π_{ref} and a non-inferiority margin δ_P, the *host-parametric HELLO scheduling problem* is

$$\min_{\pi}\ \mathbb{E}\left[B(\pi)\right] \quad \text{s.t.} \quad \mathbb{E}\left[\mathrm{PDR}(\pi)\right] \ \geq\ \mathbb{E}\left[\mathrm{PDR}(\pi_{\mathrm{ref}})\right] - \delta_P, \tag{10}$$

over causal policies $U_{j,k} = \pi_j(\mathcal{H}_j(t_k), Z_{j,k})$ that depend only on the node's own history $\mathcal{H}_j$ and local randomness $Z_{j,k}$. The problem involves the host only through the candidate set $\mathcal{I}^h$ and the score σ_i, not through its learning rule, so any host that exposes them poses the same problem. Energy enters through B and through the delivery count in (6). DA-HELLO is an experimentally assessed heuristic for (10), not an optimiser, and sending a HELLO exactly when $D_j = 1$ is not claimed to be optimal. The reference in our primary HELLO-only comparison is the 120 s periodic policy, not a policy selected for its outcome.

B. Design Overview

DA-HELLO follows three principles. It is *host-parametric*: it uses the host only through the candidate set, the score, and a broadcast primitive. It is *causal*: every input is the node's own state, a decoded frame, or a held record. And it works on *requests*: the scheduler decides when to ask for a HELLO, and the MAC decides when, and whether, it is sent. Fig. 3 shows the structure. All policies in Section V, including the no-HELLO arm, start from the same join announcement: each node sends one HELLO after a random delay, and the sink broadcasts a beacon at a fixed period throughout. After joining, DA-HELLO raises a request on four conditions.

C. Trigger Conditions

At epoch k, a node requests a HELLO if

$$U_{j,k} = \mathbf{1}\{F_j \vee M_{j,k} \vee R_{j,k} \vee \Phi_{j,k}\}, \tag{11}$$

where the conditions are evaluated in the order M, R, Φ and at most one request is raised per epoch. One HELLO serves all pending requests of the node.

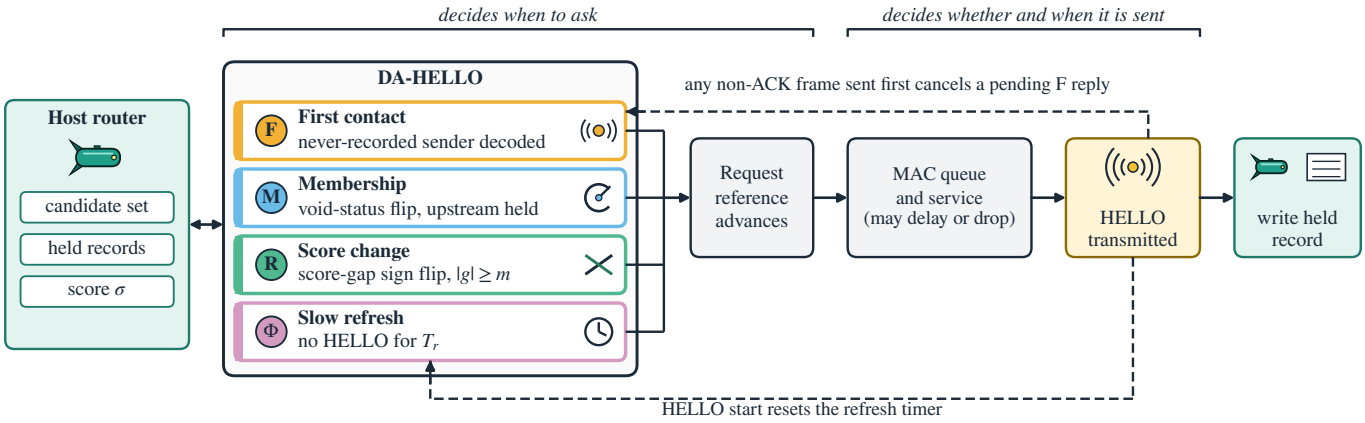


Fig. 3. Structure of DA-HELLO beside a host router. Four conditions on the node's own history raise a request; the request advances a sender reference, passes through the MAC, and a HELLO that is decoded writes the receivers' held records. A pending first-contact reply is cancelled if the node sends any non-ACK frame first.

First contact (F). When s_j decodes a frame from a sender s_x that it has never recorded, it schedules a reply after a random delay $u \sim \mathcal{U}[0, u_{\max}]$, with a minimum separation $g_{\min}$ between reply releases. Several discoveries can share one pending reply, and the separation is a release-side limit, not a bound on the spacing of physical transmissions after MAC queueing. The test uses the node's own record of s_x before the frame is written. That record is created by the first decoded state-bearing frame and is never cleared, so "new" means never recorded, not recorded and since expired. The reply is cancelled if the node begins any non-ACK transmission before it fires.

Membership (M). Let $v_j(t) = \mathbf{1}\{\mathcal{I}_j^h(t) = \emptyset\}$ be the void indicator of s_j and $\mathcal{A}_j(t)$ the approximate upstream-interest set formed from its fresh records and held depths. Then

$$M_{j,k} = \mathbf{1}\{v_j(t_k) \neq v_j^{\mathrm{ref}} \ \wedge \ \mathcal{A}_j(t_k) \neq \emptyset\}, \tag{12}$$

where v_j^{ref} is the void status at the last request. This is a void-status heuristic, not a detector for every change in a receiver's candidate set: a receiver that s_j has never heard can be absent from $\mathcal{A}_j$.

Score change (R). Let g_{jc}^{ref} and g_{jc}^{now} be the score gap between s_j and a held co-candidate s_c computed with the score at the last request and with the current score. A non-void node raises

$$R_{j,k} = \mathbf{1}\{v_j(t_k) = 0 \ \wedge \ t_k \geq \tau_j \ \wedge \ \exists\, c: \ g_{jc}^{\mathrm{ref}}\, g_{jc}^{\mathrm{now}} < 0, \ |g_{jc}^{\mathrm{now}}| \geq m\}, \tag{13}$$

with margin m and hold deadline τ_j. A void event resets the interval I_j to $I_{\min}$; each accepted rank event doubles it up to $I_{\max} = I_{\min} 2^d$ with d doublings and sets $\tau_j = t_k + I_j$. This back-off acts on rank requests only; it is distinct from MAC contention back-off and from Trickle's consistency-based suppression.

Slow refresh (Φ). With period T_r and the time t_j^{last} of the node's last HELLO,

$$\Phi_{j,k} = \mathbf{1}\{T_r < \infty \ \wedge \ t_k - t_j^{\mathrm{last}} \geq T_r\}, \tag{14}$$

and $T_r = \infty$ disables it.

Algorithm 1 DA-HELLO at node s_j

Require: held records; references g_{jc}^{ref}, v_j^{ref}, t_j^{last}, τ_j; parameters $(m, I_{\min}, d, T_r, u_{\max}, g_{\min})$

On decode of a frame from s_x (before the record is written):

if s_x never recorded by s_j **then**
  schedule or merge a first-contact reply after $u \sim \mathcal{U}[0, u_{\max}]$, not earlier than $g_{\min}$ after the previous reply release
end if

On start of any non-ACK transmission of s_j:

cancel the pending first-contact reply

At each epoch t_k:

$U \leftarrow 0$
if $M_{j,k} = 1$ **then**
  $U \leftarrow 1$; $I_j \leftarrow I_{\min}$ ▷ membership
else if $R_{j,k} = 1$ **then**
  $U \leftarrow 1$; $I_j \leftarrow \min(2I_j, \ I_{\min} 2^d)$; $\tau_j \leftarrow t_k + I_j$ ▷ score change
else if $\Phi_{j,k} = 1$ **then**
  $U \leftarrow 1$ ▷ slow refresh
end if
if $U = 1$ **then**
  update g_{jc}^{ref} and v_j^{ref} from the current state; request a HELLO
end if

D. Algorithm

Algorithm 1 gives the receive handler and the per-epoch procedure.

E. Host Instantiation

The score σ is host-specific, and each surrogate below is a local approximation of the order that neighbours compute. For MARV it is

$$\sigma_j^{\mathrm{M}} = -w_E\Big(1 - \frac{e_j}{E_0}\Big) - w_D \frac{z_j}{R_s} + \gamma(\lambda_1 - \lambda_2)V_j, \tag{15}$$

with the host's energy and depth weights w_E, w_D, normalisation energy E_0, range scale R_s, and value V_j. The current self

depth is used in both the reference and the current score, and competitor values come from the node's own held records, with the fixed sink depth taken as an a-priori constant. For QELAR it is

$$\sigma_j^{\mathrm{Q}} = \frac{e_j}{E_0} + c_n\, n_j + \gamma\, V_j, \tag{16}$$

with n_j the size of the candidate set and V_j the node's own value; the peer score uses the held energy of s_c and sets its count and value to zero. This is a heuristic proxy for the stored Q-ordering of neighbours, not an exact reconstruction, and the asymmetric substitution can influence which crossings occur. Because the rank test rests on a surrogate, a rank request is evidence for a trigger condition, not proof that a remote forwarding decision changed. The constants c_n and γ are the host's own.

F. Request Semantics of the Evaluated Implementation

Four details matter for reading the results. First, the medium samples the sender's advertised fields into a frame at the start of an actual control transmission, and receivers apply those sampled values after decoding, so a later sender value is never substituted at decode time. Second, the references advance when a request is accepted by the scheduler, before the MAC serves it, so a reference is the last requested value, not necessarily the last transmitted one; MARV also advances its sender reference on a data transmission, including in the HELLO-only variant where receivers ignore the attributes of data frames. Third, the cancellation rule is kind-based: any non-ACK transmission cancels a pending reply, whether or not it carried the fields the reply would have supplied, and in the HELLO-only variants this includes stripped data frames. Fourth, the refresh timer is reset at the start of a HELLO transmission in MARV, but at the request and at control transmissions in QELAR, so a configured T_r is not a bound on the age of a receiver's record. Counters of membership, rank, refresh, and reply-release *requests* are therefore never summed as HELLOs; physical control traffic is counted from the medium's frame-start tally. A stricter field-aware controller would change the policy, and its results would not follow from those reported here.

G. Properties and Scope

The per-epoch cost is $O(|\mathcal{N}_j|)$ in the number of held neighbours. Beyond the host primitives of Section III, DA-HELLO adds a few scalars of state per node (g_{jc}^{ref}, v_j^{ref}, t_j^{last}, τ_j) and a pending-reply timer. It is host-parametric in the sense that it needs only $\mathcal{I}^h$, σ, and a broadcast primitive. It is not self-configuring: the parameters differ by host class and state carrier (Section V).

V. Simulation Settings and Results

We evaluate DA-HELLO on two Q-learning routers, MARV [11] and QELAR [9], each in two state-carrier regimes, and report pilot runs on LQARP [13] and SAQR [10], which did not qualify as hosts. Statements, margins, arms, and seeds were fixed before the evaluation runs. All numbers below are means over seeds of run-level ratios, with ordinary 95% Student-t intervals.

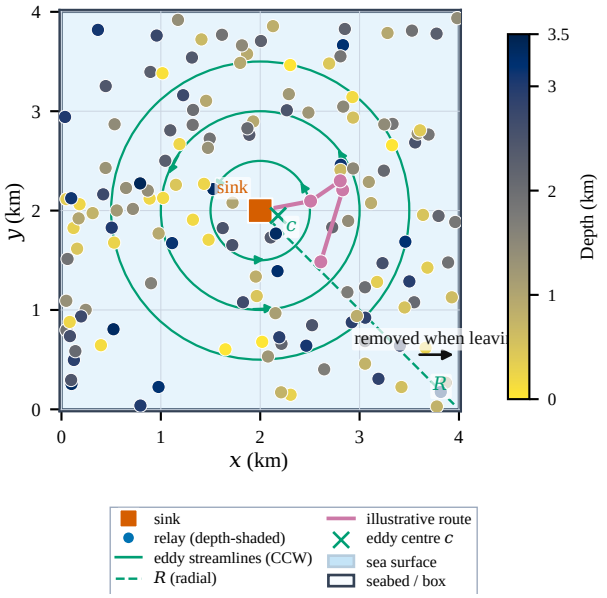


Fig. 4. Deployment used in the evaluation (one placement, seed 20261801; top view, relays shaded by depth). The sink and the eddy centre coincide. The route and the eddy streamlines are illustrative.

A. Simulation Settings

Experiments run in ns-3 with an acoustic modem and a shared-medium MAC. Table II lists the settings, and Fig. 4 shows one placement. A volume of $4000\times4000\times3500$ m holds 149 relays and a stationary sink at the centre of the horizontal plane; positions are drawn uniformly at random per seed. Every relay generates one packet on average every 1200 s, and neighbour records are created only by decoded frames (heard-only discovery) and expire after T_{nb}. Idle listening is charged for the whole run, so every energy figure includes it.

Scenarios. Scenario I is static. Scenario II applies the eddy and Gauss–Markov flow of Section III-A with a peak speed of 0.15 m/s, and Scenario GM2 doubles the speed and the variability. A relay that leaves the volume is removed, which affects on average 22.3% of the relays in II and 29.5% in GM2. We had aimed at 15% and report the measured fractions.

Intervals and estimators. The packet cohort, the energy interval, and the control count are different intervals (Section III-F): packets generated in $[900, 14400)$ s, energy in $[900, 14700]$ s, and control frames starting in $[0, 14400]$ s. The primary estimator is the mean of run-level ratios, not the pooled ratio; a pooled sensitivity analysis for QELAR gives energy differences of -181, -20, and -24 mJ/bit against 120 s periodic in Scenarios I, II, and GM2, with the same sign pattern and bootstrap intervals that exclude zero only in Scenario I.

Statistics. MARV uses evaluation seeds 20261801–40 (40 per cell; the scalability and drift sweeps use the first ten) and QELAR uses 20261841–80. DA-HELLO configurations were locked on separate tuning seeds (20261114–23) that appear in no evaluation table. Comparisons are paired by seed. Non-inferiority is stated against an explicit margin: 0.5 percentage points (pp) for MARV, and 3 pp for QELAR, because the paired standard deviation of PDR differences on QELAR was 6–10 pp in the pilot, against below 1 pp on MARV. The

TABLE II
SIMULATION SETTINGS (EVALUATION CAMPAIGNS T103/T109). SCHEDULER PARAMETERS ARE THOSE LOCKED IN SECTION V-B.

Parameter	Value	Parameter	Value
Volume / N	$4\times4\times3.5$ km / 149+1	Bit rate / HELLO, data / ACK	5 kbit/s / 400 / 64 bit
P_{tx} / P_{rx} / P_{idle}	8 / 1.3 / 0.285 W	MAC / channel / discovery	CS / formula / heard
T_{sim} / warm / drain	14400 / 900 / 300 s	Data interval / T_{nb} / beacon	1200 / 3600 / 60 s
Eddy centre / radius	(2000, 2000) / 2828 m	Sc. I / II / GM2	static / V=0.15 / V=0.3
Relay removal (II/GM2)	22% / 30% (means)	Periodic T (jittered)	{20..120} / {60, 120}
B2* Trickle	$I_{\min}$=20 s, 8 dbl., k=5	Eval. seeds (n=40)	20261801–40 / 20261841–80
Tuning seeds (pilots)	20261114–23	Non-inf. margin	0.5 pp / 3 pp (QELAR)

TABLE III
HELLO ARMS SHARED ACROSS HOSTS. ONLY THE SCHEDULER DIFFERS; JOIN ANNOUNCEMENT AND 60 S SINK BEACON ARE COMMON TO EVERY ARM.

Arm	Role	Locked settings
B0	No HELLO after join	Sink beacon only
Periodic	Jittered period T	MARV $T \in \{20, 40, 60, 120\}$; QELAR $\{60, 120\}$; free first round at t=0
Trickle	Table-change inconsistency	MARV tuned: $I_{\min}$=20 s, 8 dbl., k=5; QELAR host defaults (not matched)
DA-HELLO (locked on tuning seeds 20261114–23)		
MARV header	FC on	daMargin=0.02, daImin=10, 5 dbl., refresh 0
MARV HELLO-only	FC on	daMargin=0, daImin=2, 5 dbl., refresh 480 s
QELAR header	FC *off*	daMargin=10, daImin=10, 5 dbl., refresh 0
QELAR HELLO-only	FC on	daMargin=0, daImin=2, 5 dbl., refresh 480 s

QELAR margin was therefore chosen after the pilot and before the evaluation. Holm correction is applied within each pre-registered family of three scenarios; the intervals shown are unadjusted.

B. Hosts, Arms, and Pre-Registered Statements

Hosts. MARV and QELAR are our re-implementations of the published rules. They carry state in data headers; for each we also run a labelled HELLO-only variant in which header attributes are stripped from data frames and only HELLO or beacon decodes write records. The variants are not reproductions of a published protocol.

Arms (Table III). Every arm shares the host's routing logic and the same join announcement and 60 s sink beacon; only HELLO scheduling differs. B0 sends nothing after joining. Periodic arms send a jittered HELLO every T s and receive one free full round at time zero, which favours them; $T = 120$ s is the comparator in the pre-registered statements. The Trickle arm is tuned on MARV but uses the host's own timer defaults on QELAR, so the two Trickle results are not matched comparisons. The DA-HELLO configuration differs by host class and was locked on the tuning seeds.

Statements. S1 (MARV HELLO-only): DA-HELLO is non-inferior to periodic 120 s in PDR at 0.5 pp and uses at least 50% fewer control bytes. S1′: non-inferior in PDR, with fewer control bytes and lower energy per bit. S2 (MARV header): non-inferior to B0 at 0.5 pp with an energy change below 5%. Q1, Q1′ (QELAR HELLO-only): as S1 and S1′ at 3 pp and at least 40% fewer bytes. Q2: B0 is more than 5 pp below DA-HELLO. Q3 (QELAR header): non-inferior to the host alone at 3 pp with an energy change below 5%.

C. Results: HELLO-Only Hosts

Fig. 5 plots delivery against control overhead. Without HELLOs, both hosts collapse: B0 delivers 0.14 to 0.18 on MARV and 0.06 to 0.07 on QELAR, because no record is created after joining. Periodic HELLOs restore routing, but frequent ones hurt: on MARV, delivery in Scenario I rises from 0.45 at 20 s (376 B/s) to 0.78 at 120 s (64 B/s), since HELLO traffic congests the shared channel, and the 5 s and 10 s periods saturate it (0.01 and 0.13 in Scenario II, at 1275 and 647 B/s).

On MARV, DA-HELLO delivers 2.1 to 3.0 pp more than periodic 120 s, with 43 to 53% fewer control bytes and 8 to 10% lower energy per bit (Table IV); all intervals exclude zero. S1′ and the B0 contrast are met in every scenario, but S1 as written is met in Scenario I only, because the byte saving falls to 47.6% and 42.8% in II and GM2, below the pre-registered 50%. Against the tuned Trickle, DA-HELLO is within 0.25 to 0.60 pp of delivery, with a control rate within 4.5% and 0.9 to 1.8% lower energy per bit (the delivery interval excludes zero in II and GM2): it ties Trickle and does not beat it.

On QELAR the result is mixed. In Scenario I, DA-HELLO gains 4.95 pp over periodic 120 s with 51.5% fewer bytes and 13% lower energy per bit. In II and GM2 it is 2.2 and 1.7 pp lower, with 48% and 45% fewer bytes. The lower bounds (-4.4 and -3.7 pp) fall below the 3 pp margin, so non-inferiority is not shown, and the energy difference has an interval that spans zero. Periodic 120 s therefore keeps a small delivery edge there at about twice the overhead. Against the other comparators DA-HELLO is clearly better: relative to the host's default Trickle it delivers 3.6, 2.5, and 2.7 pp more (the interval excludes zero in I and GM2) with 33 to 36% fewer bytes and about 8% lower energy per bit, and relative to periodic 60 s it uses 72 to 76% fewer bytes and 14 to 28% less energy per bit.

Table V shows where the HELLOs come from. First-contact replies account for most of them (5055 of about 8160 starts per run on MARV and 4679 of 8094 on QELAR in Scenario II), the slow refresh for most of the rest, and membership and score-change requests together for under 160 per run.

D. Results: Hosts With State in Data Headers

When data frames already carry the state, extra HELLOs add little and cost delivery (Fig. 6). On MARV, periodic HELLOs lose 37, 33, and 30 pp at 20 s and 4.3, 2.9, and 2.9 pp at 120 s relative to B0 in Scenarios I, II, and GM2, and the tuned Trickle loses 0.4 to 0.8 pp. DA-HELLO is slightly

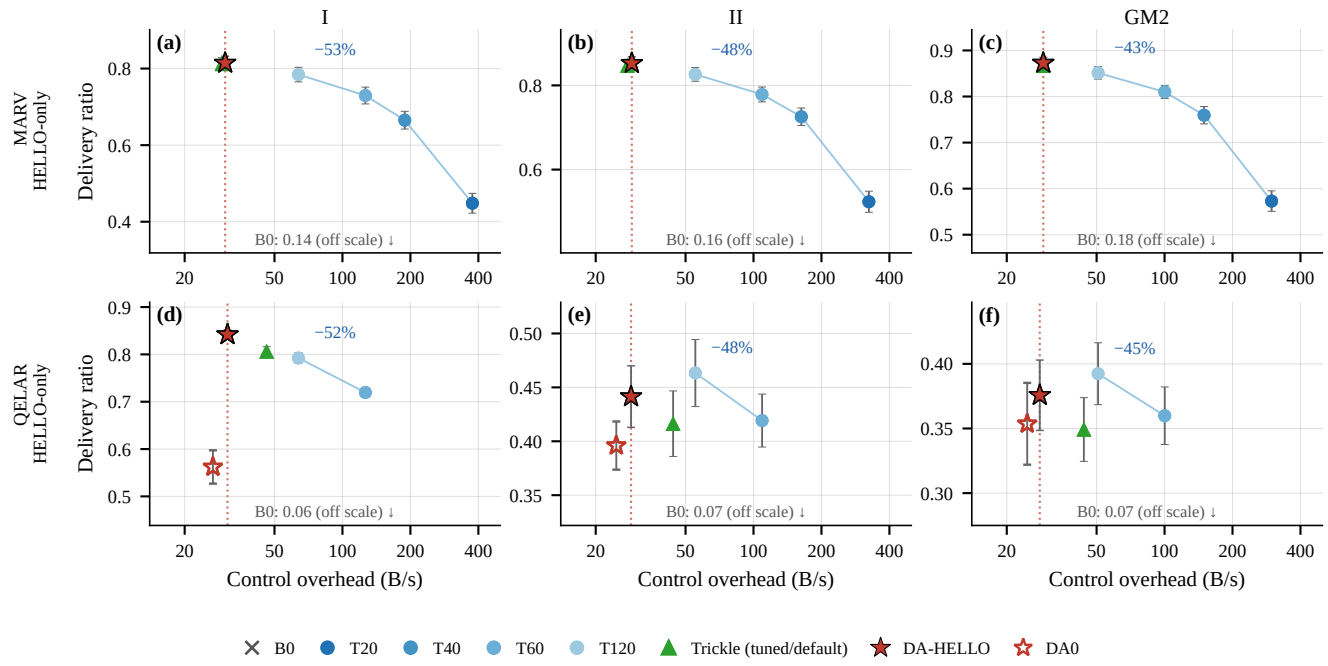


Fig. 5. Delivery versus control overhead on hosts whose state travels only in HELLOs (top: MARV variant; bottom: QELAR variant); $n = 40$ seeds, bars are 95% intervals. The no-HELLO value is off scale and printed in each panel. y-axes differ between panels. Percentages give the control bytes of DA-HELLO relative to periodic 120 s.

TABLE IV
PRE-REGISTERED COMPARISONS (MARV T103 SEEDS 20261801–40; QELAR T109 PHASE 2 SEEDS 20261841–80). Δ IS DA-HELLO MINUS THE NAMED REFERENCE; INTERVALS ARE PAIRED STUDENT-t 95%.

Stmt	Sc	ΔPDR [pp]	Rel. bytes	ΔE_{bit} [mJ/bit]
S1/S1′: MARV HELLO-only vs jittered T120				
S1	I	+3.01 [2.51, 3.51]	−52.7%	−140 [−152, −128]
	II	+2.64 [1.99, 3.29]	−47.6%	−121 [−132, −109]
	GM2	+2.15 [1.61, 2.68]	−42.8%	−104 [−114, −95]
S2: MARV header vs B0				
S2	I	+0.60 [0.01, 1.19]	—	rel −0.1% [−0.9, +0.7]
	II	+0.64 [0.09, 1.18]	—	rel −0.0% [−0.7, +0.6]
	GM2	+0.44 [0.14, 0.74]	—	rel +0.4% [+0.1, +0.7]
Q1/Q1′: QELAR HELLO-only DA480 vs T120				
Q1	I	+4.95 [4.03, 5.88]	−51.5%	−182 [−201, −163]
	II	−2.19 [−4.38, +0.00]	−48.1%	−30 [−145, +86]
	GM2	−1.67 [−3.65, +0.32]	−44.9%	+11 [−139, +161]
Q2: QELAR HELLO-only DA480 − B0				
Q2	I	+78.2 [77.3, 79.2]	—	B0 collapses
	II	+37.2 [34.5, 39.9]	—	
	GM2	+30.5 [28.1, 32.9]	—	
Q3: QELAR header (FC off, m=10) vs alone				
Q3	I	+0.16 [−0.08, +0.40]	—	rel E_{bit} hi +0.1%
	II	−0.41 [−2.42, +1.60]	—	rel E_{bit} hi +4.3%
	GM2	+0.31 [−2.03, +2.65]	—	rel E_{bit} hi +4.6%

above B0 (+0.60, +0.64, and +0.44 pp), and its energy per bit differs from B0 by at most 0.42%, so S2 is met; it also beats the tuned Trickle by 1.0 to 1.3 pp with about half the control bytes. We read this as no harm relative to doing nothing, not as a gain.

QELAR shows the same ordering with larger penalties: periodic 60 s loses 12.5, 13.7, and 14.2 pp, periodic 120 s loses 4.1, 6.3, and 7.1 pp, and the default Trickle loses 1.1, 8.0, and 11.6 pp. DA-HELLO with the first-contact reply on loses 0.2, 10.1, and 15.2 pp, and sends about 2900 first-contact HELLOs per run (Table V); with the reply off it is indistinguishable from the host alone (+0.16, −0.41, and +0.31 pp; Q3 is met) and sends about 12 membership HELLOs per run. We use the second configuration on QELAR's header regime. The two settings also differ in the rank margin, so this is a joint-configuration contrast, and we did not isolate why the reply is harmless on MARV but harmful on QELAR.

E. Components, Scalability, and Drift

Slow refresh. On the MARV HELLO-only host, removing the 480 s refresh lowers delivery by 4.4, 1.8, and 1.3 pp in Scenarios I, II, and GM2 (all intervals exclude zero) and saves 5.6 to 9.1 B/s. A 240 s refresh lowers delivery by 0.8, 0.5, and 0.5 pp relative to 480 s and costs 10 to 14 B/s more (Fig. 7). On QELAR the same removal costs 28.0, 4.5, and 2.2 pp. The refresh therefore carries most of the delivery beyond first contact on HELLO-only hosts, and it is one reason why the configuration differs between regimes.

Score-change test. The test raised between 12 and 160 host-state requests per run (Table V); on QELAR HELLO-only

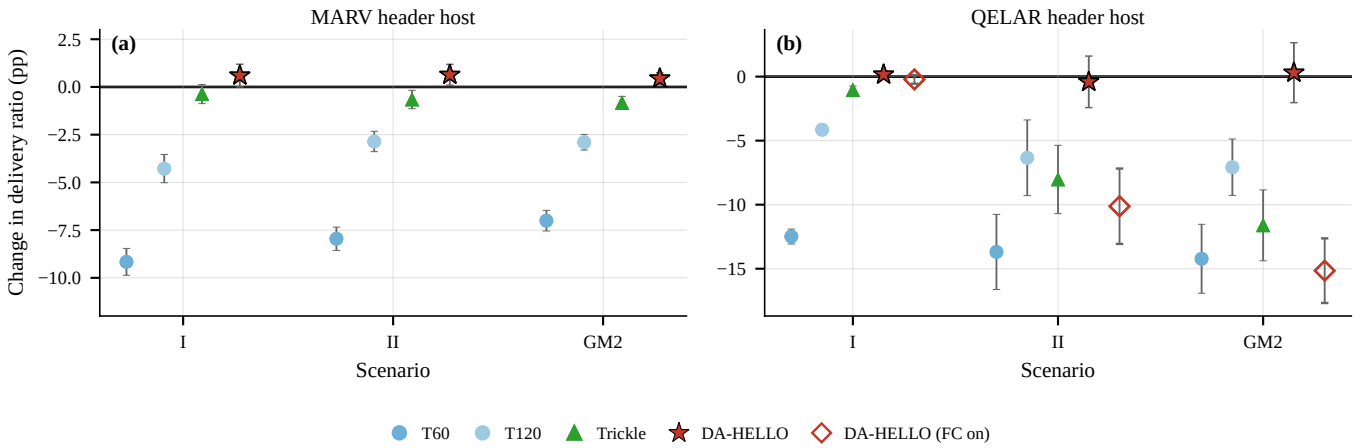


Fig. 6. Change in delivery relative to sending no HELLOs after joining (MARV) or to the host alone (QELAR), for hosts that carry state in data headers ($n = 40$ per cell, paired 95% intervals). The 20 s and 40 s MARV arms (losses of 12 to 37 pp) are omitted for scale. With first contact on, DA-HELLO on QELAR loses 10 to 15 pp in II and GM2.

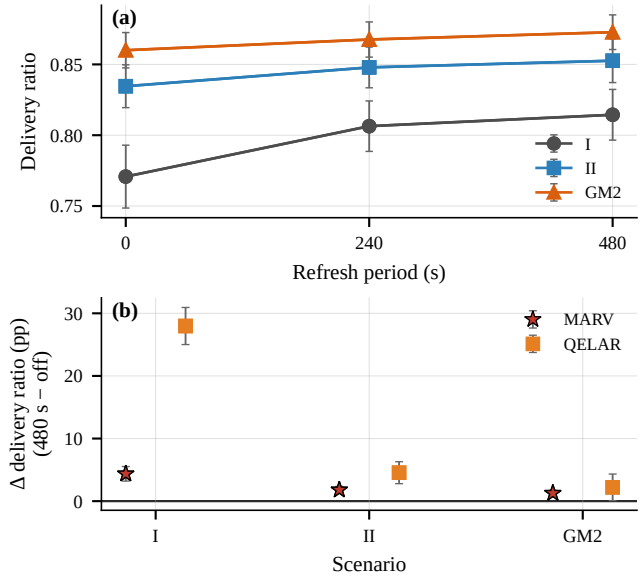


Fig. 7. Slow-refresh ablation. (a) MARV HELLO-only delivery for refresh periods of 0, 240, and 480 s. (b) Paired gain of the 480 s refresh over no refresh, for MARV and QELAR ($n = 40$, 95% intervals).

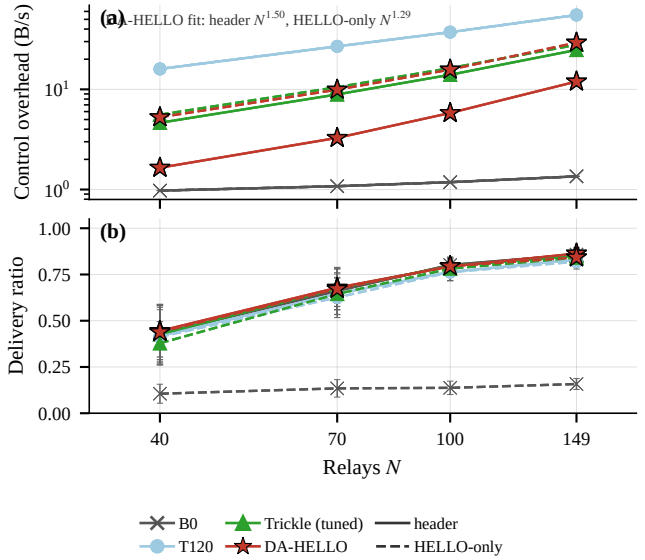


Fig. 8. Scalability in Scenario II ($n = 10$ seeds per point). Control overhead grows roughly as $N^{1.5}$ for DA-HELLO on the header host and $N^{1.3}$ on the HELLO-only host, below periodic 120 s and the tuned Trickle; delivery is low at 40 and 70 relays for all policies.

about 144 per run, roughly one per node, whereas the header configuration with the test off raised none. These are request counters, not transmissions, and we have no matched rank-on and rank-off comparison, so no performance benefit is attributed to the test or to the back-off.

Scalability. Fig. 8 varies the number of relays from 40 to 149 in Scenario II (ten seeds per point). The overhead of DA-HELLO grows from 1.7 to 12.4 B/s on the header host and from 5.1 to 29.3 B/s on the HELLO-only host, which corresponds to power laws with exponents of about 1.5 and 1.3 in N. Periodic 120 s grows from 15 to 53 B/s and the tuned Trickle from 4.8 to 26.6 B/s (header host). Delivery is low for every policy at 40 and 70 relays (0.42 to 0.44 and 0.66 to 0.69 on the header host), owing to sparse connectivity and relay removal, and DA-HELLO stays within 0.1 pp of the best comparator at 100 relays. The energy per bit at 40 relays is not reported because one zero-delivery run has no defined value.

Drift speed. Across mean speeds of 0.10, 0.15, 0.30, and 0.50 m/s (ten seeds), DA-HELLO delivers 0.839, 0.843, 0.859, and 0.867 on the HELLO-only host, against 0.813, 0.819, 0.835, and 0.843 for periodic 120 s and 0.832, 0.841, 0.852, and 0.861 for the tuned Trickle. On the header host it delivers 0.858, 0.863, 0.874, and 0.888, against 0.854, 0.858, 0.872, and 0.877 for B0.

F. Pilot Runs on Further Hosts

Before the campaign, a host had to deliver at least 0.30 of its packets in Scenario II without any HELLO scheduler on five tuning seeds. QELAR qualified (0.665 after aligning its candidate gate and previous-hop rule with its paper, up from 0.385). SAQR delivered 0.082 and LQARP 0.284 with an interval that spans the threshold, so neither was campaigned and we make no claim about DA-HELLO on them.

TABLE V
DA-HELLO HELLO STARTS BY CAUSE IN SCENARIO II ($n = 40$ SEED MEANS). FIRST-CONTACT REPLIES DOMINATE; SLOW REFRESH ACCOUNTS FOR MOST OF THE REMAINDER ON HELLO-ONLY HOSTS. MEMBERSHIP AND SCORE-CHANGE REQUESTS TOGETHER STAY UNDER 160 PER RUN.

Host / arm	HELLO tx	FC reply	Refresh	Mem./rank	Notes
MARV HELLO-only	8160	5055	2836[a]	44	refresh=480 s
QELAR HELLO-only	8094	4679	3742	158	mem 14, rank 144
QELAR header (FC off)	—	0	0	14	≈12 membership
QELAR header (FC on)	—	2907	0	141	descriptive
MARV header	3155	2981	0[a]	24	refresh=0

[a]MARV export codes slow-refresh and residual adaptive starts in `cause_other`.

G. Limitations and Disclosures

The evidence supports a bounded statement. Where state travels only in HELLOs, DA-HELLO saves 43 to 53% of the control bytes of periodic 120 s and delivers more on MARV; on QELAR it is competitive but not shown non-inferior in the drifting scenarios. Where state travels in headers, it matches doing nothing, and on QELAR only with first contact off. The configuration therefore differs by host class and is not self-adapting.

Several choices limit these results. Both hosts are our re-implementations, and 35 of 65 checklist rows of the host audit match the papers; the HELLO-only variants are constructed. Periodic arms receive a free first round, and B0 includes the sink beacon. Header bits that hosts add to data frames are not charged. The relay-removal fractions (22% and 30%) exceed the 15% target and the removal rule affects the mobile scenarios. On QELAR, the pilot suggested +2.3 and +0.5 pp against periodic 120 s, but the evaluation gave -2.2 and -1.7 pp, and the 3 pp margin was set after the pilot. A matched first-contact and rank ablation, equalised start-up for the periodic arms, and a field-aware controller remain future work. The MARV raw run logs are no longer available, so MARV results rest on the retained exports. First-contact replies have a precedent in APU [15], and the void-triggered beacon in earlier void-handling work [17].

VI. CONCLUSION

This article proposed DA-HELLO, a HELLO scheduler for Q-learning underwater routers that sends a HELLO on first contact with a new neighbour, on a change of void status, and, optionally, after a long silence. Evaluated on two re-implemented hosts, MARV and QELAR, with state carried either in data headers or only in HELLOs, the results show that the value of HELLO scheduling depends on how routing state is disseminated. Where state travels only in HELLOs, DA-HELLO delivers 2.2 to 3.0 percentage points more than a jittered 120 s periodic schedule on MARV with 43 to 53% fewer control bytes, and saves 45 to 52% of the bytes on QELAR, although non-inferiority in delivery was not established in its drifting scenarios. Where state travels in data headers, DA-HELLO matches sending no HELLOs, whereas periodic and Trickle schedules lose delivery; on QELAR this holds only with the first-contact reply switched off. The score-change test showed no demonstrated benefit. These conclusions rest on simulation of two re-implemented routers. Future work will select the configuration from the host's state carrier, test further trigger combinations, and validate against field or trace-driven current data.

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